

MOHAMED RAGAB

AI Engineer | Construction & Infrastructure

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Military status: Exempted

PROFESSIONAL SUMMARY

AI engineer with a structural engineering background, specialising in artificial intelligence for construction and infrastructure. I develop computer vision and machine learning solutions that detect structural defects in site photos, read structural drawings and convert laser scans into BIM models, taking each one from research and data preparation through model training to cloud deployment. Experienced in BIM delivery to ISO 19650 on UK projects and in structural design for US projects. MSc in Digital Construction Analytics & BIM with Distinction from Ulster University, UK.

CORE COMPETENCIES

- AI & Machine Learning
- Computer Vision & Deep Learning
- Model Training & Evaluation
- Data Engineering
- Cloud AI (RunPod, Azure)
- AI-Assisted Development
- BIM & Digital Twins
- Scan-to-BIM & CAD-to-BIM
- Structural Engineering

PROFESSIONAL EXPERIENCE

R&D AI-Construction Specialist | AGECS Apr 2026 – Present
Remote, part-time

- Develop AI models that automatically detect structural elements (columns, beams, walls, openings and piles) on structural drawings, as the foundation for converting 2D drawings into BIM models (CAD-to-BIM). The column model reaches over 93% precision and recall.
- Prepare and manage large training datasets: combining multiple public sources, cleaning and converting annotations, and quality-checking labels.
- Built a Scan-to-BIM pipeline that converts site laser scans into verified IFC models of slabs, walls and columns, automatically excluding temporary works.
- Evaluate the latest AI models, including multimodal AI, for understanding engineering drawings.

Chief Technology Officer (CTO) | AECAI Ltd Dec 2025 – Present
Belfast, UK · AI-assisted structural inspection platform

- Set the technical direction: system architecture and technology roadmap, defining how the AI models, back-end and web platform fit together.
- Make technology and vendor decisions with cost, licensing and scalability in mind, including cloud GPU infrastructure (RunPod) and commercially licensable AI models.
- Give technical sign-off on delivered work, and work with the back-end development team to integrate the AI into the platform they build and run.
- Train and deploy deep learning models that detect cracks, spalling and exposed reinforcement in inspection photos, with rigorous data and evaluation checks; improved post-processing cut false detections by two-thirds on the hardest test photo.
- Built the web platform where inspectors upload photos, the AI flags defects and engineers issue reports, using AI coding tools (Claude Code, OpenAI Codex) with code review and automated testing.
- Lead research and academic partnerships, write the technical sections of grant applications and R&D reports, and answer investors' technical questions.

Digital Design & BIM Engineer | AECOM × Ulster University Feb 2025 – May 2025
Belfast, UK · industry project

- Led a digital-twin project for a concrete water tank: built Revit models from AECOM drawings and linked them to inspection data using Dynamo and IFC.
- Developed a Power BI dashboard showing the 3D model, a crack register and recommended maintenance actions based on Eurocode 2 limits.
- Trained a deep learning crack classifier to support remote inspection. Module grade: 80%.

